

Multichannel Backscatter Communication and Ranging for Distributed Sensing With an FMCW Radar

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Abstract—This paper considers the adaptation of frequency modulated continuous wave (FMCW) radars for sensor data exfiltration in addition to their primary use in area surveillance. In particular, the application of wideband RF backscatter from semi-passive RF sensor nodes for simultaneous ranging and wireless telemetry uplink to an FMCW radar is examined. We present an upper bound on data communication rates for baseband-coded binary backscatter modulation given a particular sensor node density and radar parameters, as well as a method for line coding to mitigate stationary clutter. Proof-of-principle measurements using a brass-board S-band radar with center frequency of 2.45 GHz and 40-MHz bandwidth show the simultaneous ranging and demodulation of two sensor nodes at ranges of 15 and 33 m in a cluttered indoor environment. In this demonstration, a signaling rate of 10 kbit/s per sensor with a bit error rate below the measurable threshold of 4×10^{-3} is achieved.

Index Terms—Localization, microwave communication, radar communication, RF identification.

I. INTRODUCTION

MODULATED backscatter offers the prospect of extremely low power consumption for telemetry uplink from wireless devices [1]–[3]. Backscatter communication relies on an incident electromagnetic (EM) signal for the RF carrier and conveys information with controlled reflections of this signal [1], [4], [5]. Since backscatter devices do not need to generate their own RF carrier, they do not need to contain potentially complex and power-hungry local oscillators and transmit power amplifiers; thus enabling low-power low-cost communication links.

Backscatter communication in its simplest form is achieved with a single transistor switch that dynamically modulates the impedance presented to the sensor node's antenna between two

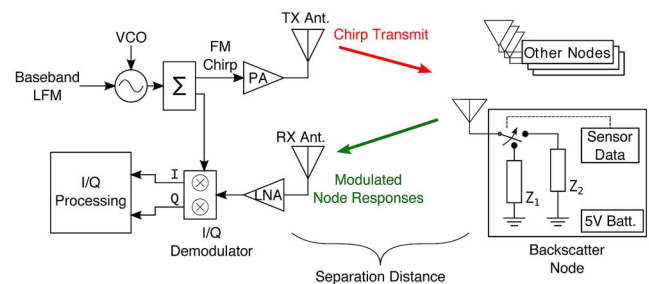


Fig. 1. Example FMCW backscatter communication system.

distinct impedances. For this reason, backscatter communication is especially well suited for sensor communication where minimizing system power and complexity are key design parameters and it has found wide deployment in the field of ultra-high-frequency (UHF) RF identification (RFID). Recent work has demonstrated the potential for backscatter-based sensors in bio-signal recording [3], [6], [7], logistics/asset monitoring [8], and environmental sensing [9].

This paper considers the application of backscatter communication for data uplink from sensors interrogated by a radar from some standoff distance, as shown in Fig. 1. In this scenario, a number of sensors are dispersed in an environment surveilled by a radar. The radar waveform is determined by the requirements of the radar's primary surveillance application, e.g., air or maritime surveillance, so it is desired that the data uplink from the sensor nodes be compatible with the existing waveform. Additionally, it is desired that range information for each sensor also be available from the radar in order to accommodate, e.g., sensors that are dispersed randomly or drift over time. This paper thus considers the problem of compatibility between the radar's ranging functionality and the added sensor uplink functionality.

Radar systems have long taken advantage of bandwidth as a resource to encode range to a target in either the time domain or frequency domain. Related work has examined the localization capabilities of linear frequency modulated (LFM) continuous wave (CW) signals with RFID tags [8], or communication for a single active RF tag [10]. In this work, the FMCW radar technique is utilized to take advantage of bandwidth for two purposes: 1) estimating the range from the radar to multiple sensors in the presence of multipath clutter and 2) separating signals from multiple sensors in view for simultaneous data uplink.

We present a processing method for separating tag data streams provided transmission rates fall within an upper bound that we derive and apply coding-based clutter filtering to aid in

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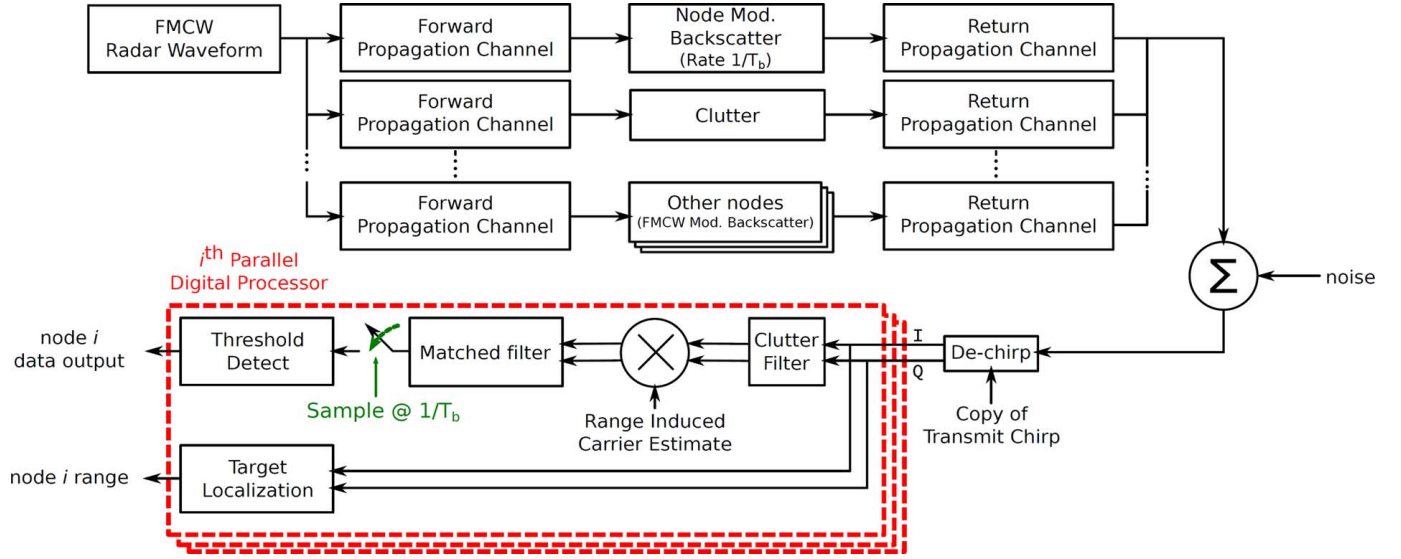


Fig. 2. Block diagram of system and signal-processing framework.

discriminating the signal of interest (SOI) from clutter returns. We consider the case where sensors do not have their own receiver, and thus cannot participate in a command-response protocol with the interrogating radar. This approach has the advantage of low complexity and reduced energy consumption at the sensors due to the elimination of receiver power as a consideration. However, it introduces significant challenges in separating responses from multiple sensors in view of the radar. This paper shows that conventional LFM waveforms can be used to separate signals from multiple sensors in view, subject to some constraints on the symbol rate and spatial proximity of the sensors.

This paper, an expanded version of [11], adds processing analysis and experimental results of multiple nodes in view, energy spillover qualitative analysis from two neighboring nodes, experimental hardware setup design details, and link budget analysis. This paper is organized as follows. Section II introduces the concept of backscatter communication with FMCW waveforms. Section III develops the necessary signal-processing framework. The tradeoffs in simultaneous node transmission with FMCW radar interrogation is discussed in Section IV. Section V develops a clutter filtering technique via line coding. The measurement results from a brass-board radar are provided in Section VI. Finally, Section VII draws a conclusion.

II. BACKSCATTER COMMUNICATION WITH FMCW WAVEFORMS

A block diagram of the proposed system is shown in Fig. 2. In this system, the radar transmits a continuous stream of linear FMCW “chirps” $x(t)$. The sweep rate $\beta = B/T_p$, where B is the chirp bandwidth and T_p is the chirp duration, determines the instantaneous bandwidth of the signal and is measured in units of Hz/s. A single chirp is described as

$$x_p(t) = e^{j2\pi f_c t + j\pi\beta t^2} \cdot \text{rect}\left[\frac{t}{T_p}\right] \quad (1)$$

where f_c is the center carrier frequency and $\text{rect}[t]$ is the rectangle function over $[-1/2, 1/2]$. The LFM waveform is then

$$x(t) = \sum_{n=-\infty}^{\infty} x_p(t - nT_p). \quad (2)$$

The backscatter modulator in the sensor node is modeled as a single switch that modulates the radar cross section (RCS) of an antenna between one of two values to generate binary coded data [12]–[14]. This work only considers semi-passive, or battery-assisted passive, sensor nodes, which use an onboard battery to supply dc power to the sensor node electronics.

In the most basic implementation, the backscatter modulator switches between two antenna loads with different impedances, i.e., open circuit ($\Gamma = +1$) and short circuit ($\Gamma = -1$), corresponding to two constellation symbol states in a binary phase shift keying (BPSK) constellation. A portion of the incident wave at the antenna is reflected by the switched load due to the load impedance mismatch and is re-radiated back to the radar [12], [15]. The switch operates at the symbol rate, not the carrier frequency, so the switching transistors need not be biased for gain at the RF carrier frequency. This minimizes power consumption in the modulator [14], [16]. The message signal driving the node switch is expressed as a series of symbols

$$m(t) = \sum_{k=-\infty}^{\infty} a_k \cdot p(t - kT_b) \quad (3)$$

where $a_k \in \{+1, -1\}$ is the binary phase shift for symbol k and $p(t)$ is a rectangular pulse of the symbol duration T_b representing a single symbol period.

The fields backscattered from node i are received at the radar with a time delay $\tau_i = 2 \cdot r_i / c$ where c is the speed of light and r_i is the range between node i and the radar.

This work assumes that the node is either stationary or moving slowly such that the Doppler frequency shift is negligible within a single symbol period. The resulting signal from

node i received at the radar is

$$s(t, r_i) = \alpha(r_i) \cdot \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \underbrace{[x_p(t - nT_p - \tau_i)]}_{\text{chirps sent by radar}} \underbrace{[a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i)]}_{\text{backscattered symbols}} \quad (4)$$

where $\alpha(r_i)$ is the attenuation due to the two-way backscatter path of length $2 \cdot r_i$ and δ_i is the delay between the radar clock and the node's clock.

The observed signal at the radar receiver also includes radar clutter, as well as multipath reflections of the backscattered signal. Radar clutter results from reflections of the transmitted chirp signal from stationary elements in the surroundings (such as terrain features, walls, buildings, etc.). The clutter can be modeled as a collection of point scatterers at different ranges so the cumulative clutter return at the radar is

$$c(t) = \sum_{n=-\infty}^{\infty} \sum_{c \in C} \alpha(r_c) x_p(t - nT_p - \tau_c) \quad (5)$$

with C representing the set of clutter points and where $\tau_c = 2r_c/c$ is the time delay associated with the range r_c of a clutter point from the radar.

Other nodes are assumed to be operating simultaneously at different ranges $\bar{r} = [r_1, \dots, r_G]$

$$S(t, \bar{r}) = \sum_{g \in G} s(t, r_g) \quad (6)$$

where G is the set of all transmitting nodes and $s(t, r_g)$ is the received signal from a node residing at range r_g , as described by (4).

The total energy at the radar receiver's input port is therefore assumed to be the sum of all the aforementioned signals (4)–(6) with additive white Gaussian noise (AWGN) $n(t)$, resulting in

$$\begin{aligned} r(t) &= s(t, r_i) + c(t) + \sum_{g \in G} s(t, r_g) + n(t) \\ &= \sum_{i \in G} \alpha(r_i) \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \left[x_p(t - nT_p - \tau_i) \cdot a_{k,i} \cdot p(t - kT_b - \frac{\tau_i}{2} - \delta_i) \right] \\ &\quad + \sum_{n=-\infty}^{\infty} \sum_{c \in C} \alpha(r_c) x_p(t - nT_p - \tau_c) + n(t) \end{aligned} \quad (7)$$

where $a_{k,i}$ is the phase shift for symbol k of node i . Multipath reflections of the backscatter signal create copies of $s(t, r_i)$ at multiple ranges resulting from the path taken between the radar and node. Potential for inter-symbol interference (ISI) is not dealt with in this paper.

III. SIGNAL-PROCESSING FRAMEWORK

A. Ranging Processing

Range localization of the node is achieved by pulse compression. In this process, the cross-correlation between the received

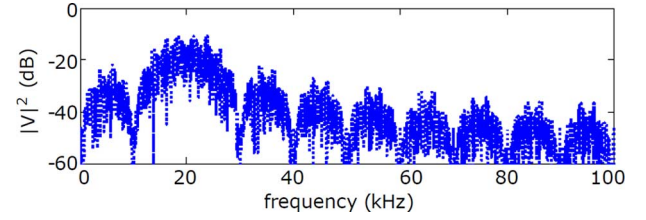


Fig. 3. Simulated received power spectrum (arbitrary power reference) of node transmitting at rate $1/T_b = 10$ kHz at range of 15 m corresponding to a range subcarrier of 20 kHz.

signal $r(t)$ and the transmitted chirp pulse signal $x(t)$ in the time domain is computed [17]

$$r_d(t) = \langle r, x \rangle(t) = \int_{t'=-\infty}^{\infty} r(t') x^*(t + t') dt'. \quad (8)$$

This correlation is typically performed as a two-step process: 1) de-chirping or de-ramping and 2) Fourier transform in fast time. The outcome of the de-chirp process is

$$\begin{aligned} r_d(t) &= \sum_{i \in G} \alpha(r_i) \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \underbrace{[a_{k,i} p(t - kT_b - \frac{\tau_i}{2} - \delta_i)]}_{\text{backscattered symbols}} \underbrace{[e^{j2\pi\beta\tau_i(t-nT_p)}]}_{\text{range induced subcarrier}} \\ &\quad + \sum_{c \in C} \alpha(r_c) \sum_{n=-\infty}^{\infty} \underbrace{[e^{j2\pi\beta\tau_c(t-nT_p)}]}_{\text{clutter point-scatterer return}} + \tilde{n}(t) \end{aligned} \quad (9)$$

where τ_i is node i delay, δ_i is delay between the radar's clock and the node's clock, and $\tilde{n}(t)$ is AWGN.

For a return at some time delay τ_i , the de-chirp processing associates a corresponding induced tone whose frequency f_i is a function of that time delay τ_i . The exact range of the node of interest can then be recovered via a discrete Fourier transform and the relation

$$f_i = \beta \cdot \tau_i = \frac{2Br_i}{cT_p}. \quad (10)$$

In the case of a single scatterer point (such as a clutter point or an inactive sensor node), a single tone will be present in the received signal after pulse compression that corresponds to the time delay. However, in the case of an actively backscattering sensor node, the signal contribution is a time-domain multiplication of the node signal $m(t)$ and the range induced subcarrier $e^{j2\pi\beta\tau_i t}$. Backscattering sensor node signals thus appear in the de-chirp outcome as a bandpass process centered around f_i , as shown in Fig. 3. Thus, the time delay τ_i leads to a range-dependent subcarrier for the backscattered signal.

To maximize range estimation accuracy, a low-rate (compared to the waveform repetition frequency) preamble signal (e.g., 101010) is pre-pended to the sensor data. This preamble concentrates the signal energy from the sensor node within a small group of range bins to enhance the separation of the node return from ground clutter. An example of a range-Doppler surface for a backscattered continuous-preamble signal is shown

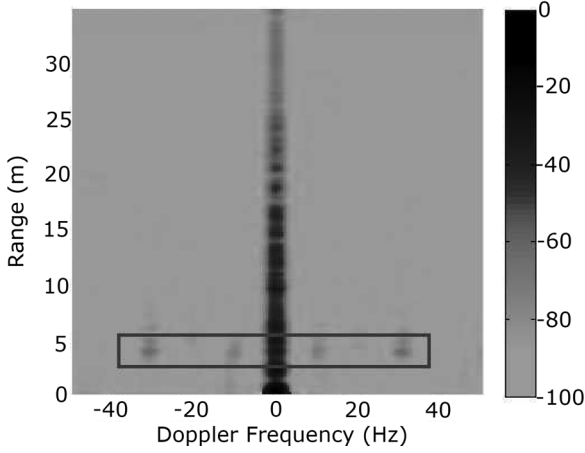


Fig. 4. Measured amplitude-range-Doppler surface plot (dB) recovered from a node at 4 m sending the preamble (1010...) at $1/T_b = 10$ Hz and a chirp rate of $1/T_p = 1$ kHz.

in Fig. 4. In this figure, the sidebands of the node's response are seen in the highlighted area. The modulation due to the preamble pushes the node response away from the zero Doppler area dominated by stationary or near-stationary clutter.

B. Communication Processing

Processing the backscattered signal from the sensor nodes includes three steps. First, a de-chirp process akin to range processing is employed to remove the FMCW modulation and obtain a baseband spectrum containing range-dependent subcarriers for each sensor node. Second, coherent demodulation using an estimated range to the sensor node is employed to demodulate the range-dependent subcarrier. Third, a matched filter is employed to recover the sensor data stream.

1) *De-Chirp Process*: To extract the sensor node uplink data, the first step is similar to range processing where de-chirping via a cross-correlation between the received and transmitted signal is performed (9). Given a backscatter return from node i at some time delay τ_i , the de-chirp processing yields a corresponding baseband tone whose frequency f_i is a function of the time delay $f(\tau_i)$. Thus, the received data uplink is

$$r_d(t) = \alpha(r_i) \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \underbrace{a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i)}_{\text{node symbols}} \cdot \underbrace{e^{j2\pi\beta\tau_i(t-nT_p)}}_{\text{range induced subcarrier}} \quad (11)$$

2) *Demodulating the Range Induced Carrier*: The second step is to demodulate each sensor node uplink from the composite received spectrum and to extract its individual spectrum. Given an initial estimate of the node range from the preamble, the corresponding subcarrier frequency f_i is estimated and used to demodulate the communications signal. The conjugated carrier replica is $e^{-j2\pi\beta\tau_q t}$ such that $\tau_q = \tau_i$. This carrier replica is mixed with the de-chirped signal. The mixing products are low-pass filtered with a cutoff at $|f| \leq 1/T_b$ to recover the

baseband signal originating from the i th node,

$$\begin{aligned} r_b(t) &= r_d(t) \cdot e^{-j2\pi\beta\tau_q(t-nT_p)} \\ &= \alpha(r_i) \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i) \\ &\quad \cdot e^{j2\pi\beta(\tau_i - \tau_q)(t-nT_p)} \\ &\stackrel{\tau_q = \tau_i}{=} \alpha(r_i) \sum_{k=-\infty}^{\infty} a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i). \end{aligned} \quad (12)$$

Range estimation error in the demodulation process has a similar impact as carrier frequency uncertainty in conventional systems [18].

3) *Matched Filtering to Recover Node Symbols*: After removal of the range induced carrier, the baseband symbols are recovered via matched filtering with an output $r_{MF}(t)$ where

$$\begin{aligned} r_{MF}(t) &= \int r_b(t-u) p^*(-u) du \\ &= \alpha(r_i) \sum_{k=-\infty}^{\infty} a_k \int p(t - kT_b - \frac{\tau_i}{2} - \delta_i - u) p^*(-u) du \\ &= \alpha(r_i) \sum_{k=-\infty}^{\infty} a_k \text{tri}\left(t - kT_b - \frac{\tau_i}{2}\right) \end{aligned} \quad (13)$$

where $p(t)$ is a rectangular pulse, and $\text{tri}(\cdot)$ is the triangle function. The phase offset δ_i between the radar and the node can be estimated from a training preamble; a variety of well-known methods for estimating this offset may be applied [18].

IV. TRADEOFFS IN SIMULTANEOUS NODE TRANSMISSIONS WITH AN FMCW RADAR

A. Upper Bound on Node Data Rate and Node Spatial Density

As described in Section III-B, the sensor node's data stream appears in the de-chirped baseband as double-sideband amplitude shift keying (ASK) or phase shift keying (PSK) modulation of a subcarrier corresponding to the distance from the radar to the node. Therefore, the signal from the i th node is centered around f_i , as shown in the spectrum of Fig. 3. In this figure, the node transmits at symbol rate $1/T_b$, which sets the occupied bandwidth.

The minimum spacing (in range) for nodes that avoids aliasing in the de-chirped baseband during simultaneous transmission is considered. We assume nodes are spaced in range by $\Delta r_{\min}$. The respective frequency spacing between range-induced subcarrier frequencies is then

$$f_1 - f_2 = \Delta f_r = \beta \Delta \tau = \frac{B}{T_p} \frac{2\Delta r_{\min}}{c}. \quad (14)$$

Assuming node 1 transmits at symbol rate $1/T_{b1}$, while node 2 transmits at symbol rate $1/T_{b2}$, to prevent spectral overlap between the two nodes the system must satisfy

$$\frac{B}{T_p} \frac{2\Delta r_{\min}}{c} \geq \left[\frac{1}{T_{b1}} + \frac{1}{T_{b2}} \right]. \quad (15)$$

This bound provides a fundamental tradeoff between node data rate, node range separation, and the bandwidth of the LFM radar waveform. In other words, uplink rate is bounded by the range

difference between simultaneously transmitting nodes. Also, as the bandwidth of the LFM radar waveform is increased, the symbol rate at which the nodes can transmit without causing spectral overlap with neighboring nodes also increases.

B. Energy Spillover With Unshaped Pulses at Minimum Separation

The backscattered signal from the nodes can be generally described as an infinite train of random symbols with a binary ASK or binary PSK modulation. The simple binary-switch modulator initially shown in Fig. 1 produces unshaped rectangular symbols due to the near-instantaneous switching of the antenna between two load impedances. Fourier analysis of these signals [19] yields a closed-form derivation of the spectrum

$$G(f) = A^2 T_b \text{sinc}^2(\pi f T_b) \quad (16)$$

where A is the pulse amplitude and T_b is the symbol duration. Approximately 90% of the signal energy is concentrated in the $\pm 1/T_b$ range around the node's subcarrier frequency. The amount of upper sideband energy in the range of $1/T_b \leq f \leq 2/T_b$ (or equivalently in the lower sideband $-2/T_b \leq f \leq -1/T_b$) is only $\approx 2.5\%$, or -16 dB relative to the in-band energy. In the case of two neighboring nodes, separated exactly by the range bound and transmitting at the same rate, the adjacent channel energy in the nearest neighbor's channel is suppressed by the same 16 dB.

V. CLUTTER FILTERING VIA LINE CODING

A. Stationary Clutter Rejection

In a typical scenario where the spatial density of sensor nodes is low, the total amount of clutter energy will far exceed the amount of energy backscattered from nodes. Clutter from a set of point scatterers manifests as a continuum of uncorrelated sinusoids in the de-chirped baseband spectrum, which in the time domain appears noise like. However, assuming the point scatterers are stationary (with zero or near-zero Doppler shift), the output of the de-chirp process due to clutter will be identical from chirp to chirp. Leveraging this observation, moving target indication (MTI) processing was developed for rejecting stationary clutter [17]. In the simplest MTI implementation, the de-chirped baseband return profile of chirp $(n-1)$ is subtracted from chirp n . Stationary clutter, which is identical from chirp to chirp, is thus rejected.

It is worth noting that under realistic conditions, as the radar pulse duration T_p increases, clutter returns from consecutive pulses may no longer be assumed to be identical. From a practical perspective, it is therefore beneficial to opt for a shorter radar pulse duration as MTI clutter mitigation is more robust.

B. Node Line Coding to Mitigate Clutter

The clutter suppression of MTI processing is leveraged to implement a simple line coding technique that allows the desired node uplink signal to be separated from the stationary clutter. In this technique, the node transmits a set of symbols corresponding to a desired message block of length M (the number

of symbols per chirp duration), which occupies the entire duration of the chirp over, for example, $0 < t \leq T_p$. On the following chirp duration (for example, $T_p < t \leq 2 \cdot T_p$), the node transmits a negated version of the same set of symbols such that $a_k = -a_{k+M}$. This coding then continues for the next pair of chirp pulses. The radar receiver decodes each pair of received baseband series by subtracting the second from the first. Stationary clutter is thus removed (as in standard MTI processing) while the decoded symbols are now enhanced as they have twice unit energy. For example, consider a trivial case of sending a message comprised of four symbols with this method,

$$\overbrace{(-1 + 1 - 1 - 1)}^{\text{first chirp}} - \overbrace{(+1 - 1 + 1 + 1)}^{\text{second chirp}} \Rightarrow \overbrace{(-2 + 2 - 2 - 2)}^{\text{decoded}}.$$

More formally, the decoding is performed on the received node signal-plus-clutter after the de-chirp process (9) and before demodulation with the range induced carrier (12),

$$r_d(t) = \sum_n \left(\alpha(r_i) \sum_{k=-\infty}^{\infty} [a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i) \cdot e^{j2\pi\beta\tau_i(t-nT_p)}] + \sum_{c \in C} \alpha(r_c) e^{j2\pi\beta\tau_c(t-nT_p)} \right). \quad (17)$$

This approach is a chirp-to-chirp coherent subtraction by processing chirp pairs yielding

$$r_{\text{decode}}(t) = \sum_{\text{odd } n} \left(\alpha(r_i) \sum_{k=-\infty}^{\infty} [2a_k p(t - kT_b - \frac{\tau_i}{2} - \delta_i)] \cdot e^{j2\pi\beta\tau_i(t-nT_p)} \right). \quad (18)$$

This line-coding approach results in each symbol of the message having twice the unit energy, while stationary clutter drops out. A simulated baseband spectrum showing this effect for message lengths $M = 2$ and $M = 8$ is provided in Fig. 5.

VI. EXPERIMENTAL RESULTS

To validate this approach, a brass-board bistatic S-band radar and accompanying backscatter sensor nodes were constructed. A block diagram of the system is shown in Fig. 6 with the digital processing blocks implementing the algorithms described above. The radar transmits at a center frequency of 2.45 GHz with a bandwidth of 40 MHz and a chirp rate of 5 kHz.

The radar's LFM waveform is generated using an Agilent N4181A RF signal generator utilizing the onboard FM option, modulated by an Agilent 33220A arbitrary waveform generator configured to generate a sawtooth linear ramp at baseband. The LFM signal is amplified using a MiniCircuits ZRL-3500+ power amplifier to a conducted power level of 26 dBm and transmitted through an L-COM HG2420EG antenna with 20-dBi gain and horizontal polarization. A photograph of this setup is shown in Fig. 7.

The semi-passive backscatter nodes were realized using a Hittite HMC241LP3 RF switch and an L-COM HG2414P 14-dBi S-band patch antenna. The baseband data applied to the RF switch is provided by another Agilent 33220A arbitrary

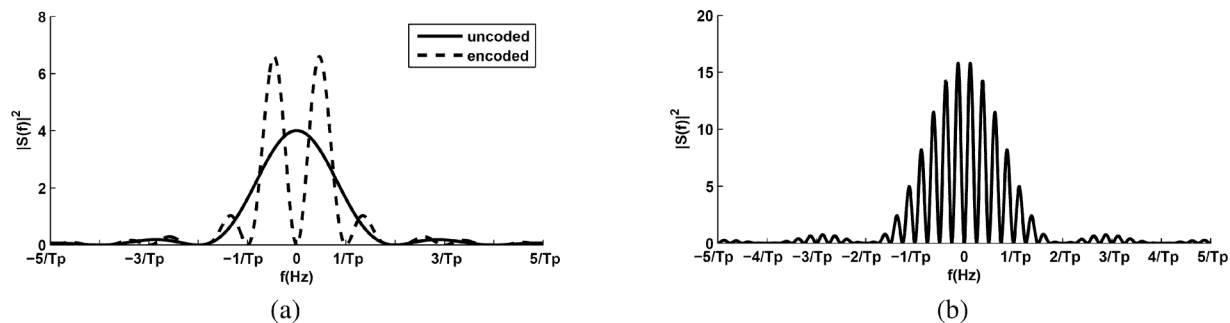


Fig. 5. Simulated baseband power spectrum density for the line coding with: (a) $M = 2$ and no coding for comparison and (b) $M = 8$.

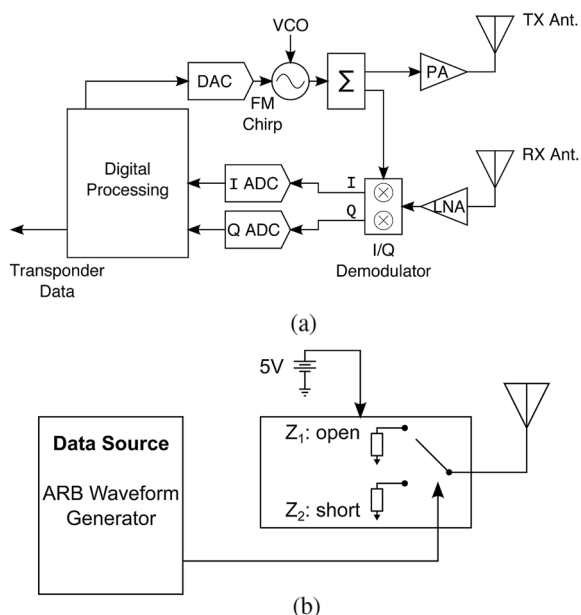


Fig. 6. Block diagram of experimental setup (analog front end). (a) Base Station. (b) Node.

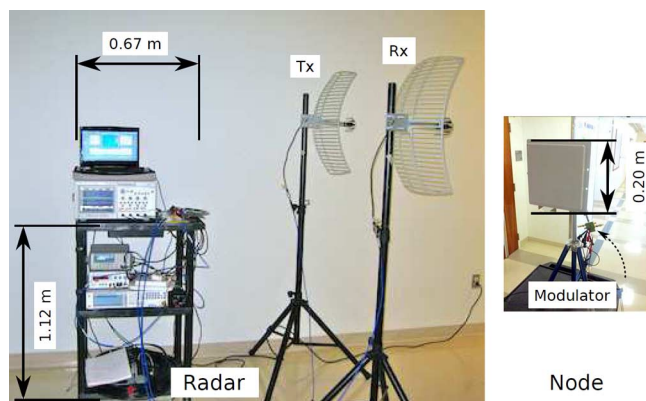


Fig. 7. Photograph of experimental radar setup in 60-m indoor hallway.

waveform generator controlled by a PC running MATLAB. This method of data generation was chosen for increased flexibility in testing the communication link. In practice, data would be passed to the modulator using a low-power microprocessor or field programmable gate array (FPGA).

For the radar's receiver, a second L-COM HG2420EG antenna is used in a bistatic configuration in conjunction with

TABLE I
EXAMPLE FMCW BACKSCATTER LINK BUDGET

Radar output power	P_t	+26 dBm
Radar antenna gain	G_t, G_r	20 dBi
Carrier wavelength (freespace)	λ	12.2 cm
Node Differential RCS	$\Delta\sigma$	4.76 dBsqm (est.)
Node Bandwidth Limit	BW	100 kHz

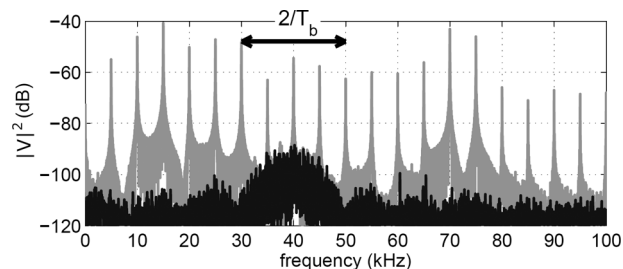


Fig. 8. Measured power spectrum of received signal before (light) and after (dark) clutter filtering with node signal centered around subcarrier of $f_s = 40$ kHz, parameters used: $f_c = 2.45$ GHz, $B = 40$ MHz, $T_p = 20$ ms, $T_b = 40$ ms.

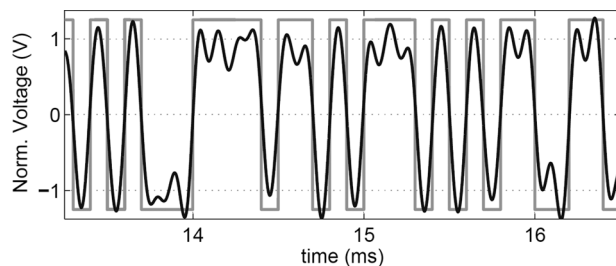


Fig. 9. Measured received signal waveform after removal of range induced carrier and clutter (dark) and transmitted signal waveform (light), parameters identical to Fig. 8.

a Mini-Circuits ZRL-3500+ RF amplifier with 20-dB gain and 2.5-dB noise figure. The received signal is de-chirped in hardware, with a copy of the transmitted signal feeding the LO port of a Linear Technologies LT5575 I/Q demodulator. The de-chirped complex baseband spectrum is passed to an Agilent DSO8104 oscilloscope, which is used for data capture. Digital processing is performed in MATLAB, as outlined in Fig. 2.

Operation of the system was first validated and characterized in an anechoic chamber and then in an indoor hallway with a maximum length of approximately 60 m. At this output power, based on the parameters in Table I, a maximum theoretical operating distance of 2 km is expected for the system to yield a

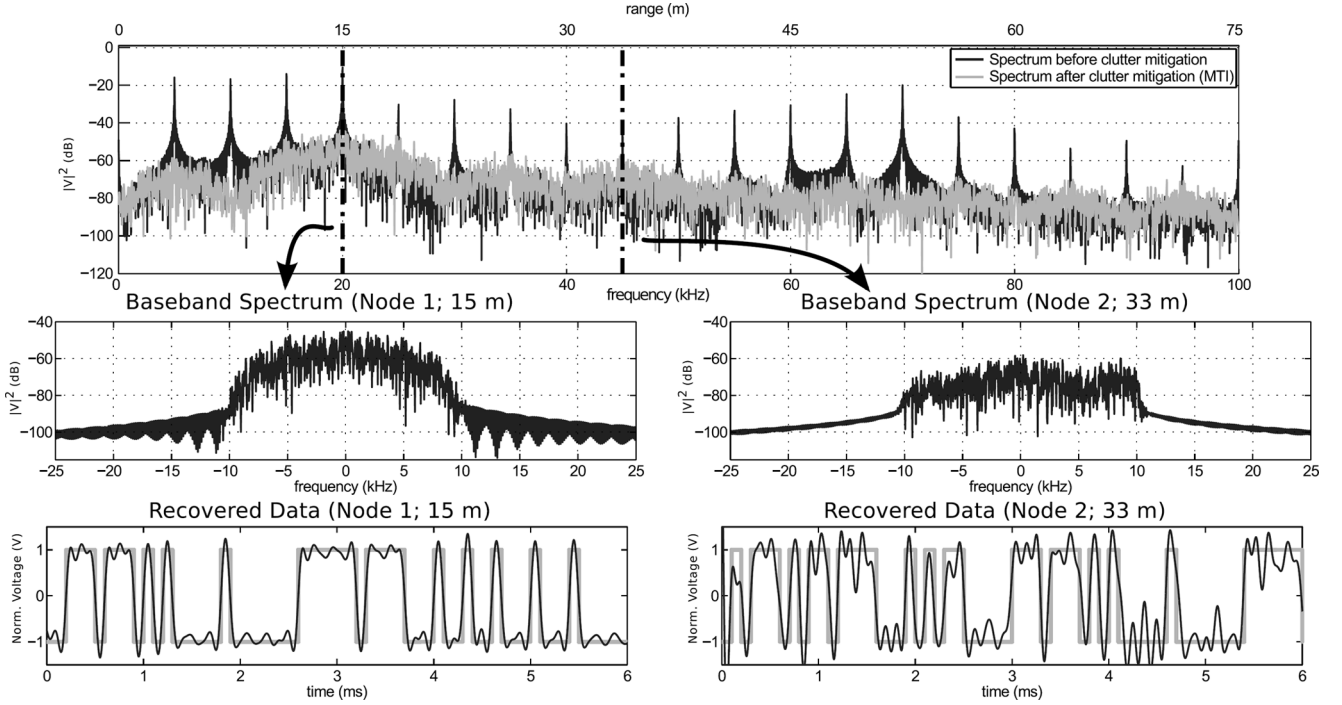


Fig. 10. Measured received power spectrum of two nodes simultaneously transmitting (*top*) filtered baseband power spectrum of nodes after range separation (*middle*) and recovered data waveform versus transmitted signal waveform prior to matched filtering (*bottom*).

signal-to-noise ratio (SNR) of 10 dB corresponding to $\leq 10^{-3}$ bit error rate (BER) for BPSK signaling from a single node. The differential radar cross section (DRCS) $\Delta\sigma$ of 4.76 dBsqm was found using the relation

$$\Delta\sigma = \frac{\lambda^2 G_{\text{ant}}^2}{4\pi} |\rho_1 - \rho_2|^2$$

where G_{ant} is the antenna gain (14 dBi), and $\rho_{1,2}$ are the power reflection coefficients [13]. In this brass-board, the range limitation is due mainly to the dynamic range of the 8-bit ADCs in the Agilent DSO8104 oscilloscope used for baseband data capture. Another limitation introduced by the use of the oscilloscope is limited baseband sample length; only 8 Mpts of data can be acquired in a single capture. One significant consequence of the limited baseband sample length is the inability to measure long runs of data; thus the BER measurements we present should be considered upper bounds on the true BER.

A. Single Node in View

Experimental results with a single node uplink at 30 m in the test hallway are shown in Fig. 8. The spectrum of the de-chirped baseband signal is shown in the light trace. The dominant components of the baseband spectrum are the peaks corresponding to stationary clutter in the hallway (multipath along the hallway), due to the periodicity of the clutter signal induced by the repetitive nature of the radar chirps. The SOI from the sensor node is masked by the clutter in the de-chirped signal, but can be recovered by differencing chirp pairs as discussed above. The resulting SOI is clearly visible in the dark trace. The time domain digital data recovered after removing the range induced subcarrier is shown in Fig. 9. The measured BER is known to be $< 4 \times 10^{-3}$ (zero errors were observed, but

limited oscilloscope sample length prevented measurements below this BER).

B. Multiple Nodes in View

Fig. 10 presents the result of an experiment demonstrating the recovery of two backscatter nodes in view simultaneously. In this experiment, each node is signaling at a rate of $1/T_b = 10$ kHz and experimental parameters are otherwise similar to previously described experiments ($f_c = 2.45$ GHz, $B = 40$ MHz, $1/T_p = 5$ kHz, $1/T_b = 10$ kHz).

Since only a single physical node was available for the experiment, the node was positioned at two separate distances and the de-chirped baseband was captured at each distance. The distances of 15 and 33 m were chosen to satisfy the range bound previously derived for the given data rate. The raw baseband signals were then summed together to generate the test signal. This quasi-realistic model is expected to be representative of the two-node case because the channel is linear and the raw complex baseband signals are added in-phase.

The top row of Fig. 10 shows the two-tag baseband power spectrum before (dark) and after (lighter) clutter mitigation and demodulation with the range-induced subcarriers for the two nodes (highlighted with dashed lines). Range to nodes were found over-the-air by examining the range-Doppler space while nodes transmitted a slow ($1/T_b < 1/T_p$) preamble (01010101 repeated). After clutter mitigation and demodulation with the two range-induced subcarrier replicas, the resulting baseband signal was then filtered using a low-pass filter with a cutoff of $f_{\text{cutoff}} = 1/T_b$ to remove out-of-channel energy. The filtered baseband power spectrum for both tags are shown in the middle row of Fig. 10. The time-domain recovered data is shown in the bottom row of Fig. 10 in black, overlaid with the transmitted

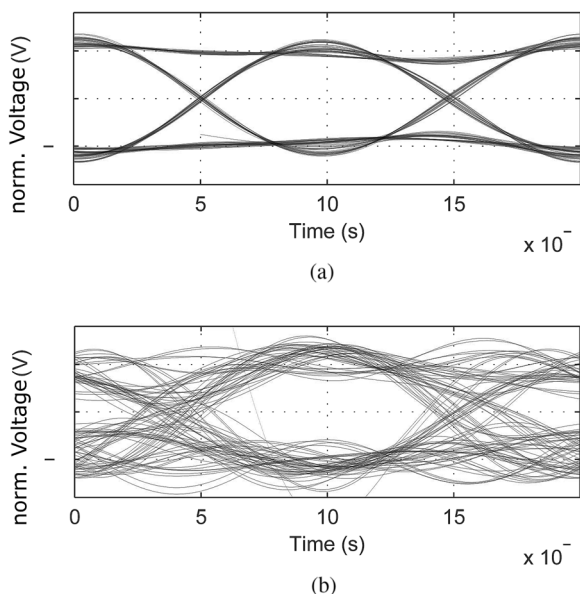


Fig. 11. Measured eye diagram for: (a) node at 15 m and (b) node at 33 m.

data in gray. Due to the increased range, the signal from the farther node (33 m) suffers from a lower SNR, but the recovered data still faithfully follows the transmitted signal. This effect can be observed more clearly in the eye diagrams for each node shown in Fig. 11. After matched filtering, all bits were compared to the transmitted bits and no bit errors were observed within the limits of the capture buffers ($\text{BER} < 4 \times 10^{-3}$).

VII. CONCLUSION

This paper has considered the adaptation of FMCW radars for sensor data exfiltration in addition to their primary use in area surveillance. In particular, it examines the application of a wideband RF backscatter from semi-passive RF sensor nodes for simultaneous ranging and wireless telemetry uplink to an FMCW radar.

Upper bounds on data communication rates for baseband-coded binary backscatter modulation, given a particular sensor node density and radar parameters, has been presented. Principally, the bound provides a fundamental tradeoff between the node data rate and the bandwidth of the LFM radar waveform forming a spectral coexistence between the radar ranging function and data communication. A method for line coding to mitigate stationary clutter is also developed.

This paper has also presented proof-of-principle measurements using a brass-board S-band (2.45 GHz) radar with a 40-MHz bandwidth, showing the simultaneous ranging and demodulation of two sensor nodes at ranges of 15 and 33 m in a cluttered indoor environment. In this demonstration, a signaling rate of 10 kbit/s per sensor at a BER of $< 4 \times 10^{-3}$ has been achieved.

Future work from a signal-processing perspective includes the development and refinement of algorithms to acquire and track node signals from a continuous real-time radar data feed.

This is expected to include significant problems in tracking clock drift among the sensor nodes, and accommodating the fact that the sensor nodes will not be synchronous to the radar's LFM chirps. Future work on the hardware includes fabrication and testing of integrated microprocessor-based S-band sensor nodes, along with the development of a near-real-time signal-processing pipeline that can be adapted to an existing radar platform. Experiments with dedicated equipment to further characterize performance of the communication link will also be performed.

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